ISSN 0020-1685, Inorganic Materials, 2011, Vol. 47, No. 3, pp. 308–312. © Pleiades Publishing, Ltd., 2011.
Original Russian Text © A.G. Belous, O.N. Gavrilenko, O.I. V'yunov, S.D. Kobilyanskaya, V.V. Trachevskii, 2011, published in Neorganicheskie Materialy, 2011, Vol. 47, No. 3, pp. 359–363.

Effect of Isovalent Substitution on the Structure and Ionic Conductivity of $Li_{0.5-y}Na_yLa_{0.5}\square Nb_2O_6$

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Received March 2, 2010; in final form, August 16, 2010

Abstract—We have synthesized $Li_{0.5-y}Na_yLa_{0.5}\square Nb_2O_6$ defect perovskite solid solutions with $0 \le y \le 0.5$. Their structure has been shown to undergo partial disordering with increasing sodium content. Lithium ion diffusion in the $Li_{0.5-y}Na_yLa_{0.5}\square Nb_2O_6$ system exhibits no percolation effects. The ionic conductivity as a function of sodium content has a maximum due to two competing factors: the increase in perovskite cell volume and the decrease in lithium ion concentration.

DOI: 10.1134/S002016851103006X

INTRODUCTION

There is considerable interest in lithium-containing multicomponent oxide systems because these can be used to produce materials with high ionic conductivity for electrochemical applications [1]. Among the best Li^+ ion conducting solid electrolytes so far are $Li_{3x}La_{2/3-x}\square_{1/3-2x}TiO_3$ defect perovskites ($\sigma \sim 10^{-3}$ S/cm at 290 K) [2–5]. The considerable vacancy concentration and high density of lithium migration channels in the structure of $Li_{3x}La_{2/3-x}\square_{4/3-2x}Nb_2O_6$ defect perovskite solid solutions enabled the fabrication of good lithium ion conductors in this system as well ($\sigma \sim 10^{-5}$ to 10^{-4} S/cm at 290 K) [6–9]. There is now substantial evidence [8, 10] that, at a sufficiently high lithium concentration in such systems, the reduction in σ is due not only to the decrease in vacancy ($\square$) concentration but also to the size effect associated with the migration channel size (the area of the narrowest section, a bottleneck, formed by four corner-shared oxygen octahedra [11]) (Fig. 1).

It is known [12] that the size of structural channels is determined by the unit-cell volume, V, which depends to a significant degree on the ionic radius on the A site of the perovskite structure. The wide range of possible isomorphous A-site substitutions makes the perovskite structure a convenient system for engineering good lithium ion conductors.

As shown earlier [13–16], partial or complete replacement of La^{3+} ($r_{[CN=12]} = 1.32$ Å) and/or Li^+ ($r_{[CN=6]} = 0.74$ Å) in perovskites by the larger sized ion Sr^{2+} ($r_{[CN=12]} = 1.44$ Å) increases to the ionic conductivity of the material.

Interesting results were obtained by substituting Na^+ ($r_{[CN=6]} = 1.02$ Å) for Li^+ in $Li_{3x}La_{2/3-x}\square_{1/3-2x}TiO_3$ [17–22]. Neutron diffraction and NMR data demonstrate that the alkali-metal and lanthanum ions in the perovskite structure of $Li_{3x-y}Na_yLa_{2/3-x}\square_{1/3-2x}TiO_3$ occupy inequivalent crystallographic sites: the Li^+ is displaced toward the center position in the faces of the unit cell, and the Na^+ and La^{+3} reside on the A site. For $y > 0.2$, lithium diffusion in this system follows a percolation mechanism [23]: at this concentration, the conductivity drops sharply because the Na^+ ions block the Li^+ migration paths.

Given that the lanthanum vacancy concentration in $Li_{3x}La_{2/3-x}\square_{4/3-2x}Nb_2O_6$ is considerably higher because the structure contains planes totally free of the large ion La^{3+} [8] (Fig. 1), it is reasonable to assume that lithium diffusion in Na-substituted lithium lanthanum niobates does not follow a percolation mechanism.

Katsumata et al. [24] and Jin Shan et al. [25] prepared and investigated $La_xNa_yLi_{1-3x-y}NbO_3$ defect per-

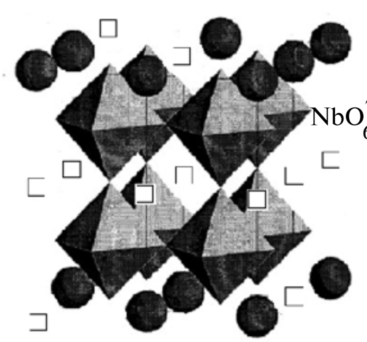

Fig. 1. Crystal structure of the $La_{2/3}\square_{4/3}Nb_2O_6$.

ovskite solid solutions. The cubic cell parameter and ionic conductivity of the solid solutions were observed to increase with Na^+ content. In those studies, however, Na^+ was substituted not only for Li^+ but also for La^{3+}. It follows from the electroneutrality condition that this should reduce the concentration of vacant sites for lithium migration. Moreover, because the ionic radius of Na^+ is smaller than that of La^{3+}, partial Na substitution for La may reduce the V of the perovskite phase. Therefore, isovalent substitution of Na^+ only for Li^+ in lithium ion conducting lanthanum niobates (with no changes in vacancy concentration) will considerably increase V and, hence, the conductivity of $Li_{0.5-y}Na_yLa_{0.5}\square Nb_2O_6$.

This paper examines the effect of isovalent substitution of Na^+ for Li^+ on the structural properties and ionic conductivity of $Li_{0.5-y}Na_yLa_{0.5}\square Nb_2O_6$ with $0 \le y \le 0.5$.

EXPERIMENTAL

$Li_{0.5-y}Na_yLa_{0.5}\square Nb_2O_6$ samples with $y = 0$, 0.1, 0.2, 0.3, 0.4, 0.43, 0.46, 0.48, and 0.5 were prepared by solid-state reactions. The starting chemicals used were LO-1 La_2O_3 and extrapure-grade Nb_2O_5, Li_2CO_3, and Na_2CO_3. The synthesis procedure was similar to that described in detail earlier [7, 8]. Appropriate powder mixtures were pressed into pellets, which were fired first at 970 K for 4 h (in order to prevent alkali-metal losses during heat treatment [8]) and then at 1320 K for 2 h, with an intermediate grinding. After homogenization by grinding in a vibratory mill with ethanol, followed by drying, an aqueous 5% solution of polyvinyl alcohol was added as a plasticizer. Green compacts ($d = 14$ mm, $p = 80$ MPa) were sintered at temperatures from 1470 to 1550 K for 2 h.

The resultant materials were characterized by X-ray diffraction (XRD). XRD patterns were collected on a DRON-4-07 powder diffractometer (CuK_α radiation). Structural parameters were determined by the Rietveld profile analysis method using XRD data.

In electrical measurements, we used samples 12 mm in diameter and 1 mm in thickness. Pt electrodes (0.5 μm) were deposited by electron-beam evaporation. The impedance of our samples was measured from 100 Hz to 1 MHz with a Solartron Analytical 1260A impedance/gain-phase analyzer. The electrical equivalent circuit and its components were identified using the Frequency Response Analyser 4.7 program.

^7Li nuclear magnetic resonance (NMR) spectra were measured on a Bruker Avance 400 spectrometer at frequencies of 155.51 and 105.84 MHz and temperatures from 235 to 350 K. The chemical shift was determined relative to $Li(H_2O)_4^+Cl^-$. The NMR profile function parameters (broad Gaussian and narrow Lorentzian components) were determined using the PeakFit program.

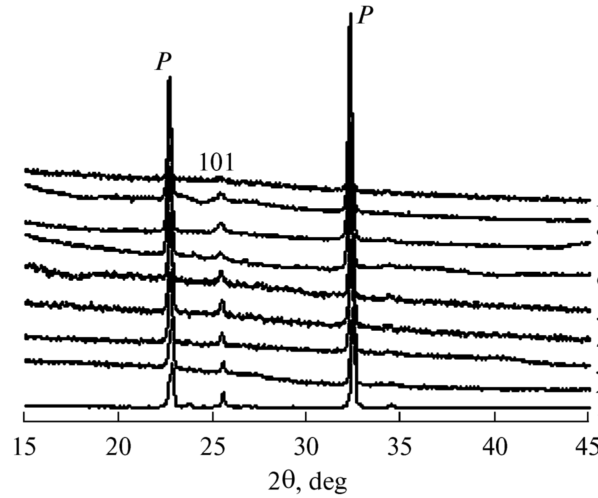

Fig. 2. XRD patterns of sintered $Li_{0.5-y}Na_yLa_{0.5}\square Nb_2O_6$ samples with $y =$ (*1*) 0, (*2*) 0.1, (*3*) 0.2, (*4*) 0.3, (*5*) 0.4, (*6*) 0.43, (*7*) 0.46, (*8*) 0.48, and (*9*) 0.5.

RESULTS AND DISCUSSION

XRD examination (Fig. 2) showed that, regardless of the sodium content, the sintered materials were single-phase and had an orthorhombic defect perovskite structure (sp. gr. *Pmmm*). Over the entire range of isovalent substitutions studied, we observed the 101 superlattice reflection, attesting to additional vacancy ordering [26] in the $z = 1/2$ plane (Fig. 1). Its intensity decreases with increasing y, which may be interpreted as evidence that sodium ions substitute for lithium ions in both the $z = 0$ and $z = 1/2$ planes, giving rise to structural disordering.

The table presents the structural parameters of $Li_{0.5-y}Na_yLa_{0.5}\square Nb_2O_6$ multicomponent oxides differing in sodium content. The positional parameters in the structure of $Li_{3x}La_{2/3-x}\square_{4/3-2x}Nb_2O_6$ [26] were used as input parameters. The unit-cell volume increases with sodium content, following Vegard's law (table), because the ionic radius of sodium is greater than that of lithium.

Because the large ion Na^+ is incapable of participating in ionic transport in such systems [17–22], the ionic conductivity of $Li_{0.5-y}Na_yLa_{0.5}\square Nb_2O_6$ is determined by Li^+ transport. Figure 3 presents the complex impedance data for our samples. The complex impedance spectra have the form of a portion of a semicircle, representing ion relaxation in the grain bulk and at grain boundaries, and a straight line, due to electrode polarization. As seen in Fig. 3, sodium substitution for lithium reduces the resistance of the our samples, with a minimum at $y = 0.43$. Further increase in Na^+ content leads to a rise in resistance. From the impedance data, we obtained the composition dependences of the total ionic conductivity (bulk and grain-boundary components) shown in Fig. 4. Sodium substitution for lithium increases the 290-K ionic con-

Unit-cell parameters, atomic position coordinates, and agreement factors for $Li_{0.5-y}Na_yLa_{0.5}\square Nb_2O_6$

y	0.1	0.2	0.3	0.4	0.43	0.46	0.48	0.5
a, Å	3.903(8)	3.906(8)	3.915(1)	3.915(9)	3.923(1)	3.918(1)	3.9167(8)	3.925(1)
b, Å	3.904(7)	3.907(8)	3.912(1)	3.915(9)	3.9311(6)	3.923(1)	3.9304(7)	3.9278(6)
c, Å	7.854(2)	7.852(1)	7.861(1)	7.861(1)	7.850(2)	7.860(2)	7.846(2)	7.848(3)
V, Å^3	119.7(3)	119.8(4)	120.40(5)	120.5(2)	121.07(5)	120.79(6)	120.78(4)	120.98(6)
Nb	0.254(2)	0.256(1)	0.256(1)	0.259(1)	0.2527(8)	0.2551(9)	0.2582(9)	0.251(1)
O3	0.24(2)	0.30(4)	0.26(2)	0.3(3)	0.28(1)	0.26(4)	0.260(6)	0.29(4)
O4	0.21(2)	0.28(4)	0.29(2)	0.3(3)	0.29(1)	0.26(4)	0.272(7)	0.28(4)
R_f, %	6.42	6.98	5.92	5.36	7.36	6.30	5.94	6.40
R_{exp}, %	8.62	9.15	9.78	8.36	5.01	4.84	4.18	8.96

Note: Positions of atoms and vacancies: La (1*a*), 0 0 0; Nb (2*t*), 1/2 1/2 *z*; O1 (1*f*), 1/2 1/2 0; O2 (1*h*), 1/2 1/2 1/2; O3 (2*s*), 1/2 0 *z*; O4 (2*r*), 0 1/2 *z*; (1*c*), 0 0 1/2.

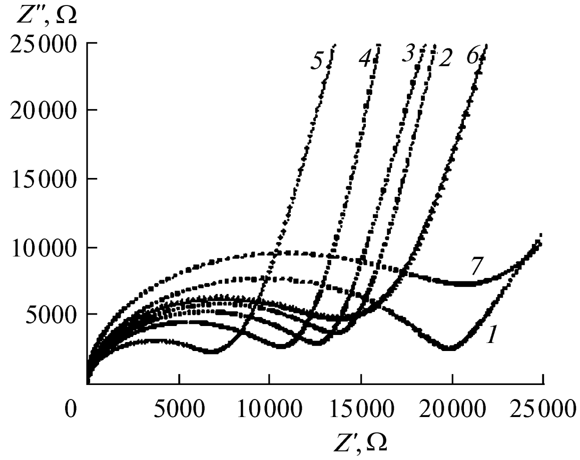

Fig. 3. 290-K complex impedance spectra of the $Li_{0.5-y}Na_yLa_{0.5}\square Nb_2O_6$ samples with y = (*1*) 0, (*2*) 0.1, (*3*) 0.2, (*4*) 0.3, (*5*) 0.43, (*6*) 0.46, and (*7*) 0.48.

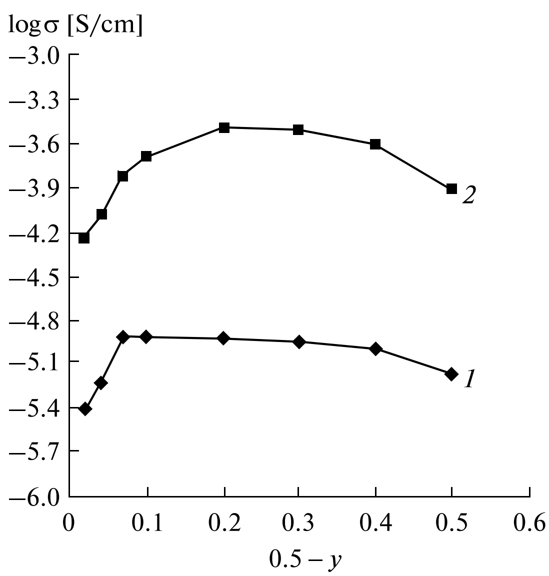

Fig. 4. Conductivity as a function of lithium content for $Li_{0.5-y}Na_yLa_{0.5}\square Nb_2O_6$ at (*1*) 290 and (*2*) 370 K.

ductivity of $Li_{0.5-y}Na_yLa_{0.5}\square Nb_2O_6$ from $\sigma = 6.85 \times 10^{-6}$ S/cm at $y = 0$ to 1.28×10^{-5} S/cm at $y = 0.43$ (Fig. 4). Further increase in y sharply reduces σ. Moreover, raising the temperature from 290 to 370 K shifts the maximum in σ to lower Na^+ contents ($y \sim 0.3$).

Since the ^{7}Li nucleus ($I = 3/2$) has a quadrupole moment, ^{7}Li NMR parameters depend on the interaction of the quadrupole moment of the nucleus with an electric field gradient. We detected no pronounced quadrupole interaction (satellites) in the ^{7}Li NMR spectra [10] of $Li_{0.5-y}Na_yLa_{0.5}\square Nb_2O_6$ (Fig. 5), which might be caused by the insufficient lithium ion mobility in the temperature range studied. The chemical shift, δ, of resonances depended on y (Fig. 5a) and temperature (Fig. 5b), and the signal became narrower with increasing temperature (Fig. 5b). The NMR spectra of our samples comprise a broad (I_b) and a narrow (I_n) component, which are due to the presence of lithium ions differing in nearest neighbor environment (lithium sites in the $z = 0$ and $z = 1/2$ planes (Fig. 1)) and mobility [8, 17–21]. The broad component is due to the less mobile lithium ions, and the narrow one, to the more mobile lithium. The decrease in I_b/I_n ratio upon sodium substitution for lithium (Fig. 5c) points to an increase in lithium mobility, which may be due to the increase in unit-cell volume. The downfield shift at $y = 0.48$ (Fig. 5a) is attributable to the decrease in ionic conductivity at this Na^+ content, accompanied by weaker shielding of the lithium nucleus. The upfield shift with increasing temperature at $y = 0.43$ (Fig. 5b) indicates that the field of the nearest neighbor ions has a stronger effect at higher σ values. The narrowing of the signal is also related to the increase in lithium ion conductivity with temperature.

The ionic conductivity of such systems is known to have a maximum at a certain migration channel size [4, 5]. The $Li_{3x}La_{2/3-x}\square_{4/3-2x}Nb_2O_6$ system was shown to have a maximum in ionic conductivity as a function of lithium concentration [7, 8].

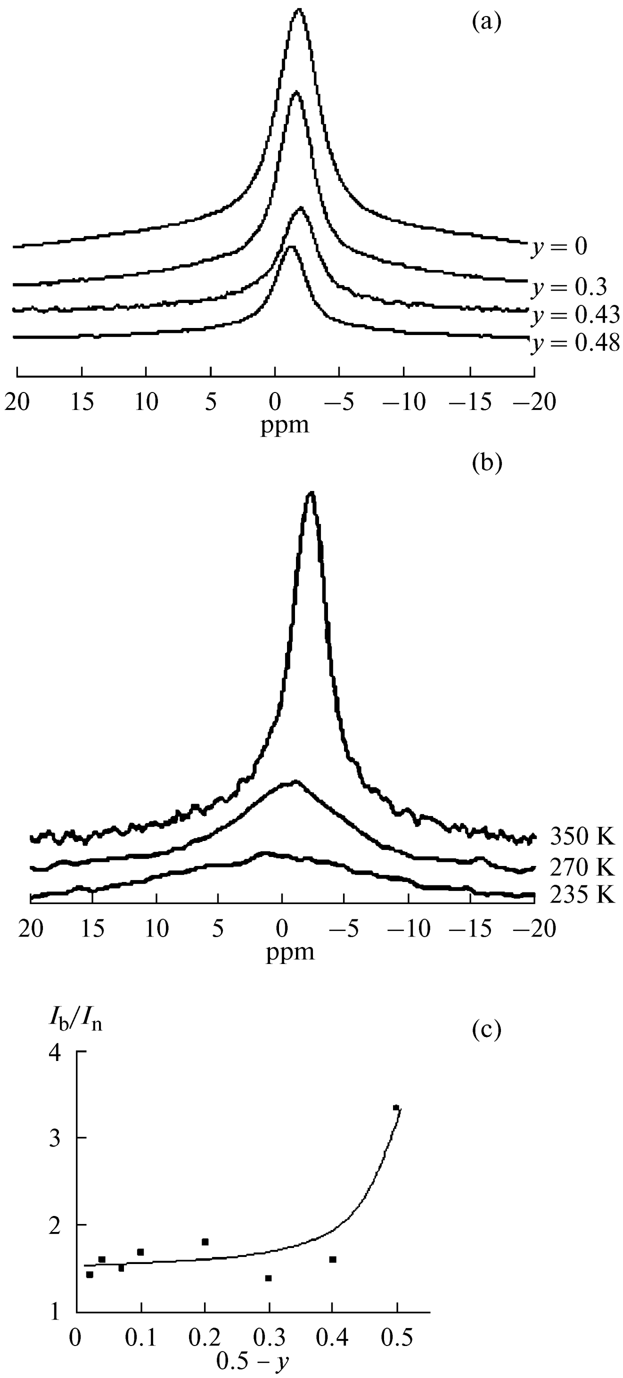

Fig. 5. ^{7}Li NMR spectra of the $Li_{0.5-y}Na_y La_{0.5}\square Nb_2O$ samples: (a) influence of y at 350 K, (b) influence of temperature at $y = 0.43$. (c) Intensity ratio of the broad and narrow components as a function of lithium content at 300 K.

Analysis of the data in Figs. 3 and 4 leads us to conclude that the ionic conductivity of the $Li_{0.5-y}Na_yLa_{0.5}\square Nb_2O_6$ samples is influenced by two competing factors, depending on y. On the one hand, an increase in y leads to an increase in V and, accordingly, in ionic conductivity. On the other hand, increasing the Na^+ concentration reduces the lithium concentration in the system and, hence, the conductivity. The competition between these effects results in a maximum in lithium ion conductivity as a function of y.

CONCLUSIONS

The present results demonstrate that $Li_{0.5-y}Na_yLa_{0.5}\square Nb_2O_6$ solid solutions with $0 \le y \le 0.5$ have a defect perovskite structure (orthorhombic symmetry, sp. gr. *Pmmm*). With increasing sodium content, the structure undergoes partial disordering. Lithium ion diffusion in the $Li_{0.5-y}Na_yLa_{0.5}\square Nb_2O_6$ system exhibits no percolation effects. The ionic conductivity as a function of sodium content has a maximum due to two competing factors: the increase in perovskite cell volume and the decrease in lithium ion concentration.

REFERENCES

1. Burmakin, E.I., *Tverdye elektrolity s provodimost'yu po kationam shchelochnykh metallov* (Alkali Metal Ion Conducting Solid Electrolytes), Moscow: Nauka, 1992, p. 263.
2. Belous, A.G., Butko, V.I., Novitskaya, G.N., et al., Electrical Conductivity of $La_{2/3-x}M_{3x}TiO_3$ Perovskites, *Ukr. Fiz. Zh.* (Russ. Ed.), 1986, vol. 31, no. 4, pp. 576–581.
3. Belous, A.G., Properties of Heterosubstituted Titanates with Perovskite Structure, *3rd Euro-Ceramics*, 1993, vol. 2, pp. 341–346.
4. Inaguma, Y., Liquan, C., Itoh, M., et al., High Ionic Conductivity in Lithium Lanthanum Titanate, *Solid State Commun.,* 1993, vol. 86, no. 10, pp. 689–695.
5. Ibarra, J., Varez, A., Leon, C., et al., Influence of Composition on the Structure and Conductivity of the Fast Ionic Conductors $La_{2/3-x}Li_{3x}TiO_3$ $(0.03 \le x \le 0.167)$, *Solid State Ionics*, 2000, vol. 1, pp. 1–9.
6. Belous, A.G., Didukh, I.R., Novosadova, E.B., and Pashkova, E.V., Electrical Conductivity of $(Pb_{1-x}La_{2/3-x-y}M_{3x})B_2O_6$, *Fiz. Tverd. Tela* (Leningrad), 1986, vol. 28, no. 10, pp. 3232–3236.
7. Belous, A.G., Gavrilenko, O.N., Pashkova, E.V., and Mirnyi, V.N., Lithium Ion Conductivity and Crystal-Chemical Aspects of $La_{2/3-x}Li_{3x}\square_{4/3-2x}Nb_2O_6$ Defect Perovskite Solid Solutions, *Elektrokhimiya*, 2002, vol. 38, no. 4, pp. 479–484.
8. Belous, A., Pashkova, E., Gavrilenko, O., et al., Solid Electrolytes Based on Lithium-Containing Lanthanum Metaniobates, *J. Eur. Ceram. Soc.*, 2004, vol. 24, pp. 1301–1304.
9. Belous, A.G., Gavrilenko, O.N., Pashkova, E.V., et al., Effect of Synthesis Conditions on the Lithium Nonstoichiometry and Properties of $La_{2/3-x}Li_{3x}\square_{4/3-2x}M_2O_6$ (M = Nb, Ta) Perovskite-like Solid Solutions, *Neorg. Mater.*, 2004, vol. 40, no. 8, pp. 993–1000 [*Inorg. Mater.* (Engl. Transl.), vol. 40, no. 8, pp. 867–873].
10. Paris, M.A., Sanz, J., Leon, C., et al., Li Mobility in the Orthorhombic $Li_{0.18}La_{0.61}TiO_3$ Perovskite Studied by NMR and Impedance Spectroscopies, *Chem. Mater.*, 2000, vol. 12, pp. 1694–1701.

11. Maier, J., Nano-Sized Mixed Conductors (Aspects of Nano-Ionics. Part III), *Solid State Ionics*, 2002, vol. 148, pp. 367–374.

12. West, A.R., *Solid State Chemistry and Its Applications*, Chichester: Wiley, 1985. Translated under the title *Khimiya tverdogo tela*, Moscow: Mir, 1988, pp. 440–441.

13. Itoh, M., Inaguma, Y., Jung, W., et al., High Lithium Ion Conductivity in the Perovskite-Type Compounds $Ln_{1/2}Li_{1/2}TiO_3$ (Ln = La, Pr, Nd, Sm), *Solis State Ionics*, 1995, vols. 70–71, pp. 203–207.

14. Thangadura, V.I., Shukla, A.K., and Gopalakrishnn, J., $LiSr_{1.650.35}B_{1.3}B'_{1.7}O_9$(B = Ti, Zr; B' = Nb, Ta): New Lithium Ion Conductors Based on the Perovskite Structure, *Chem. Mater.*, 1999, vol. 11, pp. 835–839.

15. Watanabe, H. and Kuwano, J., Formation of Perovskite Solid Solutions and Lithium-Ion Conductivity in the Compositions $Li_{2x}Sr_{1-2x}M^{III}_{0.5-x}Ta_{0.5+x}O_3$ (M = Cr, Fe, Co, Al, Ga, In, Y), *J. Power Sources*, 1997, vol. 68, pp. 421–426.

16. Gavrilenko, O.N., Belous, A.G., Kovalenko, L.L., and Pashkova, Ye.V., Effect of the A-Site Substitution on the Structure Peculiarities and Ionic Conductivity of Solid Electrolytes $La_{2/3-x-y}Li_{3x-y}Sr_{2y}\square_{4/3-2x}Nb_2O_6$, *Mater. Manuf. Processes*, 2008, vol. 23, pp. 607–610.

17. Rivera, A., León C., Santamaria, J., et al., Percolation-Limited Ionic Diffusion in $Li_{0.5-x}Na_xLa_{0.5}TiO_3$ Perovskites ($0 < x < 0.5$), *Chem. Mater.*, 2002, vol. 14, no. 12, pp. 5148–5152.

18. Sanz, J., Alonso, J.,Várez, A., et al., Structural Analysis of Li-Ion Conducting Perovskites $Li_{0.5-x}Na_xLa_{0.5}TiO_3$, *The Institute Laue-Langevin Annual Report*, 2002, pp. 34–35.

19. Sanz, J., Rivera, A., León, C., et al., Li Mobility in $(Li, Na)_yLa_{0.66-y/3}TiO_3$ Perovskites ($0.09 < y < 0.5$). A Model System for the Percolation Theory, *Mater. Res. Soc. Proc.*, 2003, vol. 756, pp. EE2.31–EE2.36.

20. Herrero, C.,Várez, A., Rivera, A., et al., Influence of Vacancy Ordering on the Percolative Behavior of $(Li_{1-x}Na_x)_{3y}La_{2/3y}TiO_3$ Perovskites, *J. Phys. Chem. B*, 2005, vol. 109, no. 8, pp. 3262–3268.

21. Sanjuan, M., Laguna, M., Belous, A., and V'yunov, O., On the Local Structure and Lithium Dynamics of $La_{0.5}(Li, Na)_{0.5}TiO_3$ Ionic Conductors. A Raman Study, *Chem. Mater.*, 2005, vol. 17, no. 23, pp. 5862–5866.

22. Jimene, R., Rivera, A., Várez, A., and Sanz, J., Li Mobility in $Li_{0.5-x}Na_xLa_{0.5}TiO_3$ Perovskites (0×0.5), *Solid State Ionics*, 2009, vol. 180, no. 26, pp. 1362–1371.

23. Stauffer, D. and Aharony, A., *Introduction to Percolation Theory*, London: Taylor and Francis, 1992.

24. Katsumata, T., Inaguma, Y., and Itoh, M., New Perovskite-Type Lithium Ion Conductors, $La_xM_yLi_{1-3x-y}NbO_3$ (M = Ag and Na), *Solid State Ionics,* 1998, vols. 113–115, pp. 465–469.

25. Jin Shan, Y., Sinozaki, N., and Nakamura, T., Preparation and Characterizations of New Perovskite Oxides $La_xNa_{1-3x-y}Li_y\square_{2x}NbO_3$ ($0 \le x$ and $y \le 0.2$), *Solid State Ionics*, 1998, vol. 108, pp. 403–406.

26. Nalbandyan, V.B. and Shukaev, I.A., Novel Tantalates and Niobates, *Zh. Neorg. Khim*, 1989, vol. 34, pp. 793–795.